

Rohit Singh

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Professional Summary

Software Engineer with 4+ years of experience designing and shipping backend systems, distributed data platforms, and serverless APIs. Strong expertise in **Go and Python** with proven experience building **high-throughput services** (sub-millisecond lookups across 2B+ records), leading platform modernization efforts, and delivering security-critical tooling under production incidents. Proven ownership from system design through production deployment.

Experience

Senior Associate Software Engineer, SonicWall – Remote

Aug 2023 – Present

- **Telemetry Ingestion Service** (Go · Microservices)
Built a **two-service streaming pipeline** (ingestor + store) in Go for collecting firewall telemetry from millions of devices, using **zero-buffer multipart streaming** and spool-based async storage with drain workers for crash-safe, backpressure-tolerant ingestion. Deployed as independently scalable containers with per-IP rate limiting, GeoIP resolution, and full CI/CD.
- **Credential Auditor API** (Go · AWS · Bloom Filters)
Built a **serverless breach detection system** processing **2B+ credential hashes** with Bloom filters for **sub-millisecond lookups**, served through a dual-mode API (encrypted + pre-hashed) with **mTLS authentication** and 16K-record bulk batches. Deployed on a containerized CI/CD pipeline with auto-scaling.
- **Distributed Telemetry Processing Platform** (FastAPI · Django · RabbitMQ)
Built a distributed telemetry platform adopted across security, product, and engineering teams to run **data-driven analysis** over thousands of firewall telemetry files, through a self-service web UI. Decoupled the **FastAPI** processing backend from the **Django** frontend so workers scale independently, with a persistent callback-and-retry layer guaranteeing result delivery through network failures.
- **Firewall Configuration Migration Platform** (FastAPI · React · Docker)
Led a full-stack rewrite of a decade-old **Perl/CGI** firewall migration tool into a modern **FastAPI and React** application, supporting migration across 80+ device models and 3 firmware generations. Replaced a brittle single-page form with a guided four-phase wizard adding firmware-compatibility validation, capacity checks, and audit logging.
- **Security Incident Response Tool** (Go · WASM)
Delivered a **zero-dependency Go + WASM** tool during a live security incident to strip sensitive data (VPN keys, credentials, RADIUS secrets) from firewall configurations, with an offline-capable browser runtime and a CLI remediation generator. **Recognized with spot award by CEO.**
- **Firewall Diagnostics & Analysis Platform** (Python · FastAPI · Vue)
Refactored a monolithic firewall diagnostics tool into a modular three-tier platform that parses device telemetry and auto-surfaces configuration risks, firmware issues, and policy anomalies for support teams. Introduced a **rule engine** to decouple detection logic from core code, making analysis rules easier to add, test, and maintain.

Associate Software Engineer, SonicWall – Bengaluru

May 2022 – Aug 2023

- Maintained and extended enterprise migration tools for firmware compatibility across new device generations.
- Built automation utilities that reduced manual onboarding effort and improved system reliability.

Software Developer, Auton Systems Pvt. Ltd – Maharashtra

Sep 2021 – May 2022

- Developed IoT and embedded systems using Raspberry Pi and ESP32.
- Worked on hardware–software integration for sensor-based systems and automation workflows.

Skills

Languages	Go, Python, JavaScript, TypeScript, C, Bash, Rust
Backend & API	Go (net/http), FastAPI, Django, Celery, REST APIs, WASM, Pydantic
Distributed Systems	Microservices, Event-Driven Architecture, RabbitMQ, Message Queues, Concurrency, Observability, System Design
Databases & DS	SQL (SQLite, PostgreSQL), NoSQL (MongoDB, Redis, Memcached), Bloom Filters
Cloud	AWS (ECS, Fargate, Lambda, KMS, SSM, ALB, ECR), Terraform
DevOps	Docker, Docker Compose, Nginx, GitLab CI/CD, Linux/Unix
Networking	TCP/IP, HTTP/HTTPS, DNS, IP Subnetting, NAT, Routing
Security	mTLS, Cryptography (AES/DES), OWASP, SAML, LDAP
Frontend	React, Vue.js, TypeScript, Tailwind CSS, Zustand, TanStack Query
AI Tooling	Claude Code, LLM-assisted debugging/development

Education

Mumbai University, B.Sc. in Computer Science — CGPA: 8.0/10.0	2018 – 2021
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Certifications & Achievements

- **Spot Award by CEO** (Oct 2025) — Critical security tool delivery during company-wide incident
- **Certificate of Recognition** (Jan 2024) — Self-initiated platform architecture refactoring
- Developing Serverless Solutions on AWS | Building Data Lakes on AWS | AWS Well-Architected Best Practices